

RoboSmart Investment — Project Summary

Course: AI & Innovation in Capital Markets · **Track 3 — Software Product**
Deliverable: Streamlit web application, deployed to Hugging Face Spaces
Figures in this document were recorded on 2026-07-26 against live market data.

1. The financial problem

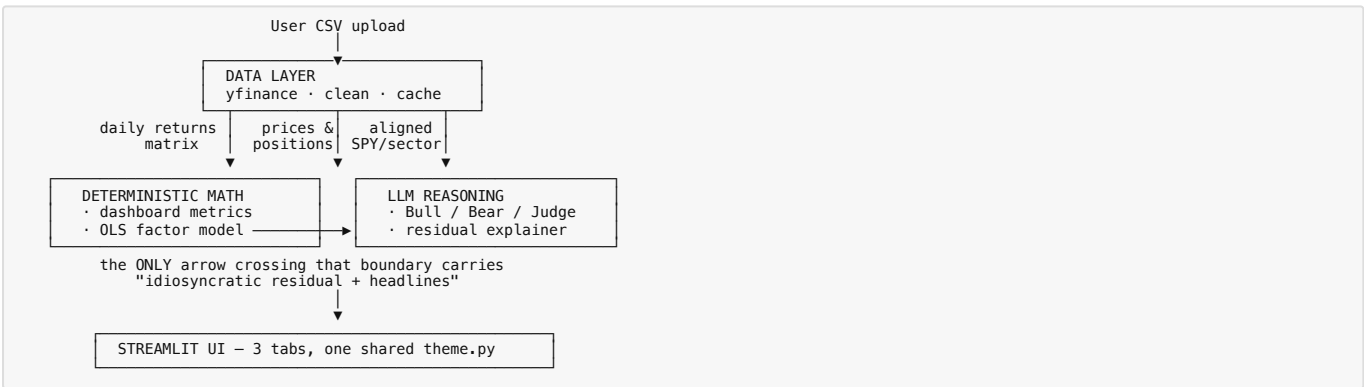
A retail investor trying to understand their own portfolio has to choose between two tools that fail in opposite directions.

Brokerage dashboards and spreadsheet exports are **rigorous but mute**. They report what a portfolio is worth and whether it is up or down. They never interpret. They do not say “this is dangerously concentrated,” or “most of your daily moves are the market moving and have nothing to do with the companies you picked,” or “today’s drop is not explained by anything in the news, so be careful attributing a story to it.” The numbers are correct and the meaning is absent — and meaning is exactly what the investor lacks.

The new generation of AI stock assistants is **fluent but ungrounded**. Ask a general-purpose language model why a stock moved and it returns a confident paragraph that is often disconnected from the actual price action, invents a catalyst, or dresses up a market-wide move as company news. It reads like analysis with no measured quantity behind it, no decomposition, and no acknowledgement of uncertainty.

RoboSmart occupies the gap. The design premise is that **every qualitative statement the system makes is anchored to a quantity the system computed**, and that the quantitative machinery is deterministic, inspectable and unit-tested rather than generated by a model. The language model is used only where it adds value: turning pre-computed numbers into readable reasoning, and interpreting the one residual quantity that genuinely requires external text. The honesty requirement is central — the system is engineered to be *able* to say “inconclusive” or “no clear cause found,” because a tool that is always confident is a tool that is sometimes confidently wrong.

2. Architecture



The single crossing arrow is the design. The language model never sees a raw price series and never computes a number. It receives one figure the regression already produced, and its only job is to say what might explain it.

Keeping deterministic math and LLM reasoning in physically separate modules is not stylistic. First, the numbers are never at the mercy of a stochastic model — a beta is a regression coefficient, not a token the model happened to emit. Second, it makes the system testable where testing is meaningful: you cannot unit-test a paragraph of prose, but you can unit-test that HHI, beta, R², drawdown and the factor decomposition are correct. **38 tests cover the quantitative layer.** Third, it localises risk: the failure modes of a language model are quarantined to the two places where the reader knows to treat output as argument rather than fact.

Runtime modes. Market data is always fetched live from `yfinance`. `USE MOCK=1` substitutes *recorded AI output* for the two language-model features, so the app is fully demonstrable with no API key and at zero cost — the mode is labelled in the UI, and recorded debates are banner-marked with the ticker they were recorded for.

3. Method

3.1 The factor model, and why the sector ETF is residualized

Each holding’s daily return is modelled as

$$r_{\text{stock}} = \alpha + \beta_{\text{mkt}} \cdot r_{\text{SPY}} + \beta_{\text{sector}} \cdot r_{\text{SECTOR_RESID}} + \varepsilon$$

The residualization step is the methodological core. A sector ETF is itself heavily correlated with the broad market: when SPY rises, a technology ETF almost always rises too. Regressing the stock on *raw* SPY and *raw* sector returns makes the two regressors collinear, so the market's influence is double-counted and split arbitrarily between the betas. The fix is a two-stage (Frisch–Waugh–Lovell) construction — first regress the sector ETF on SPY, $r_{\text{SECTOR}} = a + b \cdot r_{\text{SPY}} + u$, and keep the residual u , which is the part of the sector's movement *orthogonal to the market*: sector news net of the market. Because u is uncorrelated with r_{SPY} by construction, β_{mkt} cleanly captures market exposure, β_{sector} captures pure sector exposure, and the contributions add up without overlap. Only ϵ — the company-specific piece — is handed to the language model.

Worked example (NVDA, 2026-07-26): total -0.92% = market $+0.19\%$ + sector -1.48% + idiosyncratic $+0.37\%$, with $R^2 = 54\%$, $\beta_{\text{mkt}} = 1.87$, $\beta_{\text{sector}} = 0.91$ over 250 days. The market was mildly up; the market-neutral semiconductor sector was sharply down and did nearly all the damage; something company-specific pushed *back up*. Note the residual points the **opposite way** to the headline number — precisely what a raw “NVDA fell today” reading hides.

3.2 The multi-agent debate, and the prompt rules that enforce it

Three agents argue over one holding across five rounds: a **Bull**, a **Bear**, and a **Judge**. The design is deliberately adversarial and anti-sycophantic, and each choice is enforced in the system prompts rather than left to chance:

- **Evidence grounding.** Every claim must cite a specific number from the data (a beta, correlation, weight, drawdown, P&L figure) or a supplied headline. Assertions without a cited quantity are disallowed. This is the mechanism that keeps the debate out of the ungrounded failure mode of §1.
- **Forced disagreement.** The Bear may *never* concede — a guard against the documented tendency of language models to converge into bland consensus. Forbidding capitulation guarantees the strongest counter-argument is always articulated.
- **A Judge that may abstain.** The Judge is explicitly permitted to return “**inconclusive**.” A tool forced to pick a winner manufactures false confidence; allowing abstention is what makes a decisive verdict worth something.
- **Mandatory falsifiers.** The Judge must output **three concrete, checkable falsifiers** — “what would change my mind.” This converts a verdict from an opinion into a testable position and hands the user a watch-list rather than a conclusion to take on faith.
- **Permitted “I don’t know.”** Both the Judge and the residual explainer are allowed and encouraged to decline. Making abstention a first-class, sanctioned output is the single most important honesty lever in the system.

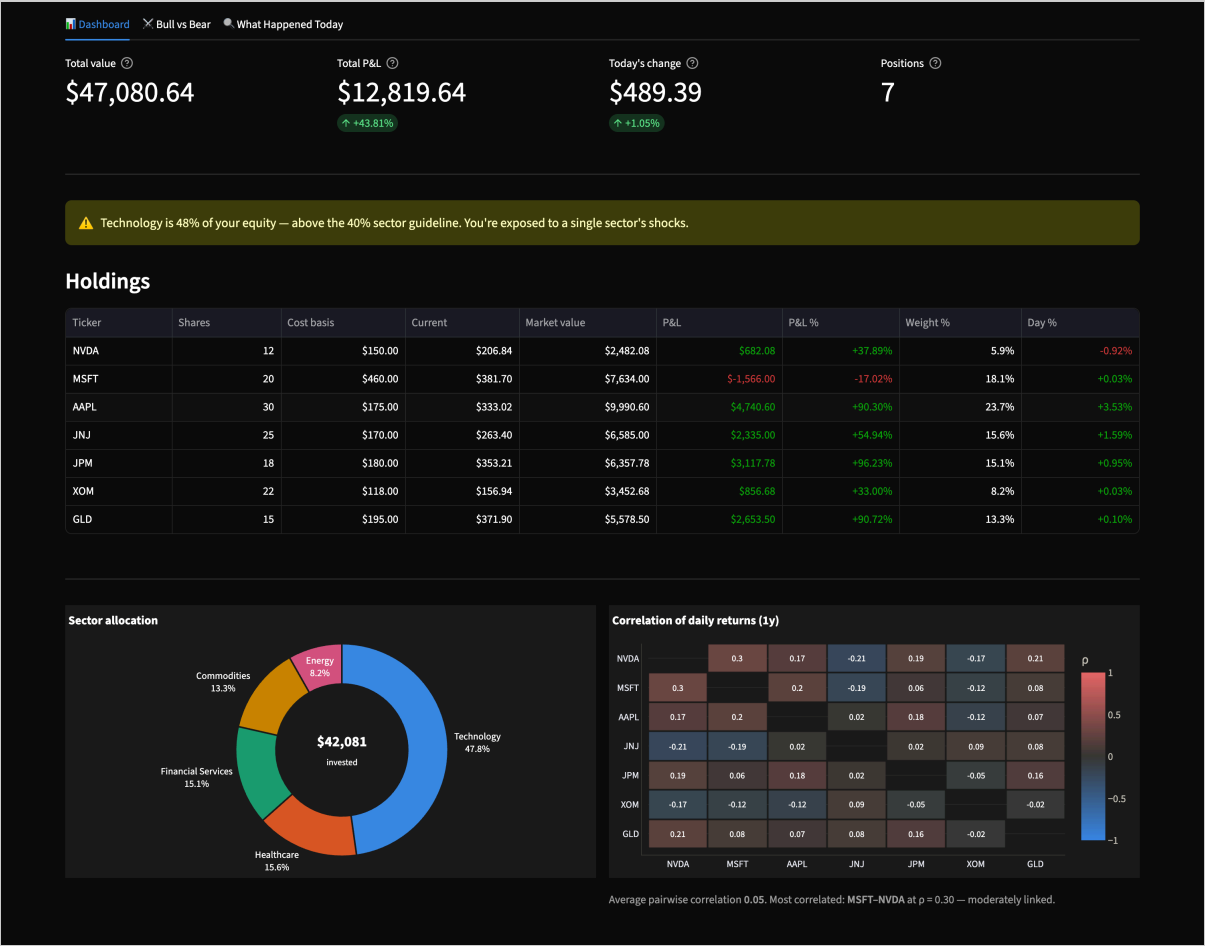
3.3 Dashboard methods

- **Portfolio beta with R^2 .** Beta is estimated by regressing the *portfolio’s* own daily return series on the market’s, and **R^2 is reported beside it** so the reader can judge how much the single market factor actually explains. (*Demo book: $\beta = 0.60$, $R^2 = 43\%$ on 250 days.*)
- **Correlation on daily returns, not price levels** — correlations of levels are dominated by common trends and are largely spurious. (*Demo book: average pairwise correlation 0.05; most-correlated pair MSFT–NVDA at $\rho = 0.30$; JNJ negative to both and GLD lowest in the book. Verified against an independent computation, agreeing to three decimals.*)
- **Concentration thresholds** — warnings when a single position exceeds 25%, any sector exceeds 40%, or the top three together exceed 60%. (*Demo book: Technology at 48% trips the sector flag.*)
- **Effective number of holdings** — inverse Herfindahl index $1/\Sigma w_i^2$, answering “how many equally-weighted positions would give the same concentration?” (*Demo book: 6.1 of 7 held.*)
- **Risk metrics** — annualised volatility (daily $\sigma \times \sqrt{252}$) and maximum drawdown (largest peak-to-trough decline of the cumulative path), plus cumulative performance against the S&P 500 with both series indexed to 100.

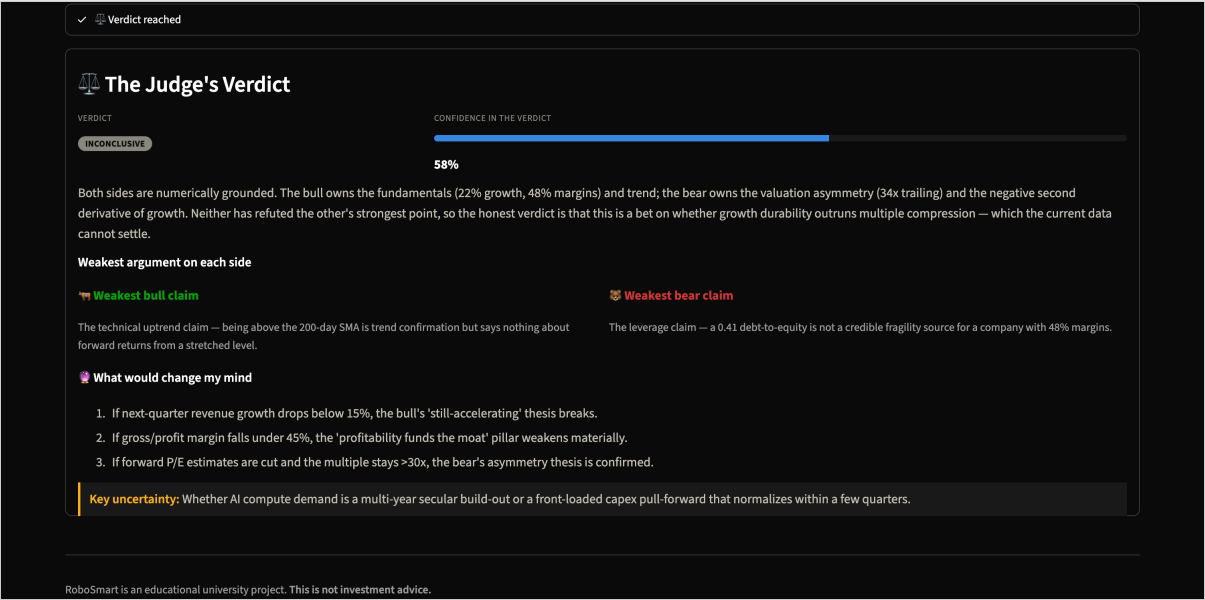
4. Results

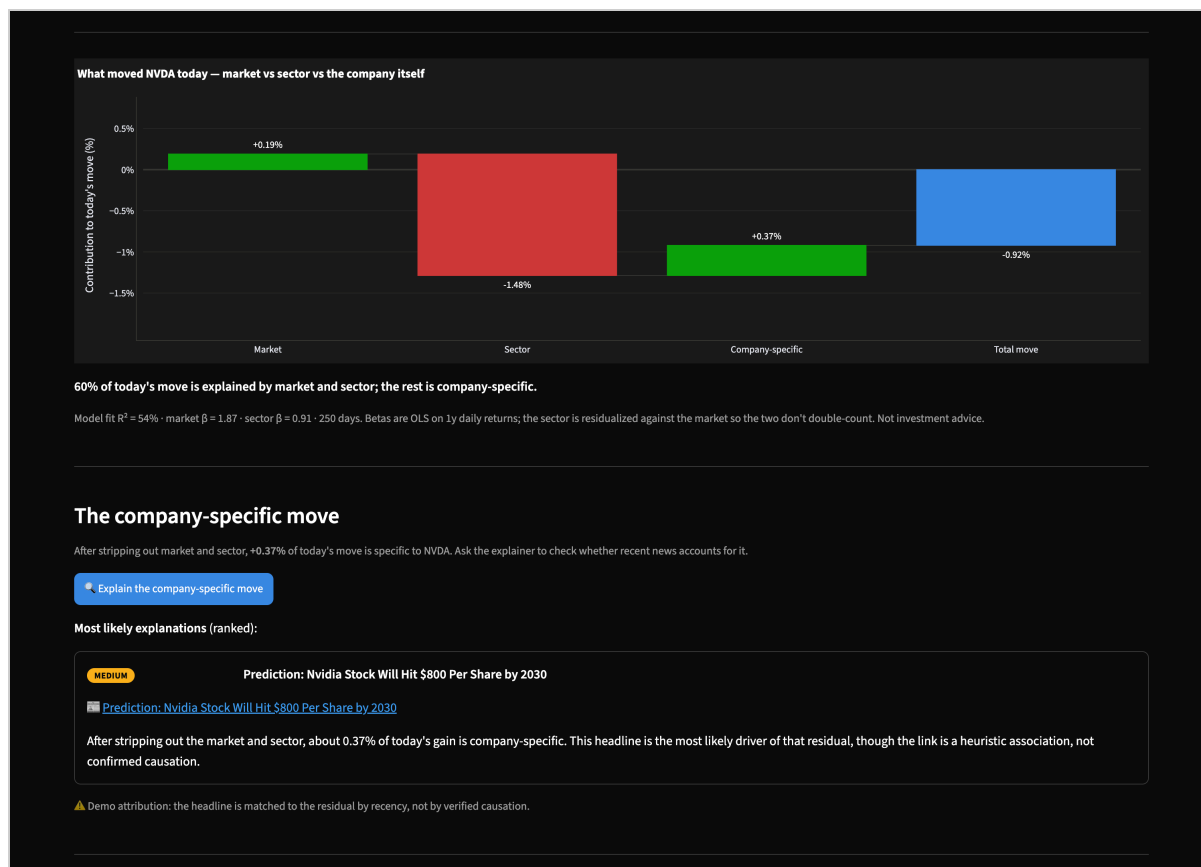
A three-tab Streamlit application that runs end-to-end on live market data, with 38 passing tests over the quantitative layer.

Portfolio Dashboard. On the bundled demo book (2026-07-26): $\beta = 0.60$ with $R^2 = 43\%$, average pairwise correlation **0.05**, **6.1 effective holdings** of 7, a **Technology = 48%** concentration flag, annualised volatility **11.8%**, maximum drawdown -7.4% , and a 1-year return of **$+28.2\%$ ($+10.9\%$ vs the S&P 500).**



Bull vs Bear. A five-round debate streams in; Bull and Bear open with number-cited arguments, the Bear presses without conceding, and the Judge closes with a verdict — here **inconclusive at 58% confidence** — plus its three falsifiers.





5. Risks & Caveats

- **The news feed was mostly about other companies — measured, then mitigated.** We assumed the limitation was thin, lagging coverage. Measuring it showed something worse: of 70 headlines yfinance attached to our seven holdings, only **29 (41%) mentioned the company they were attached to**. NVDA's top story was about Teva Pharmaceutical. This mattered because those headlines are injected into the debate agents' system prompt and the explainer's evidence list, and both prompts instruct the model to cite a headline — so the feed was actively inviting the fabrication the prompts exist to prevent. Yahoo exposes two news endpoints and neither is clean alone: `Ticker().news` is a generic “related stories” panel; `Search()` is better on some names (NVDA 8/10) and worse on others (XOM 3/10); and its `relatedTickers` field is untrustworthy — Yahoo tags “Dividend ETFs vs. Bond ETFs” as NVDA. We therefore pool both sources, de-duplicate, and keep only headlines naming the company, raising relevance to **95%** at the cost of volume (≈ 5.7 headlines per ticker instead of 10). The residual limitation is narrower but real: retail news APIs rank by engagement, not by relevance to a ticker, and a precision filter cannot manufacture coverage that was never published.
- **Betas are unstable and regime-dependent.** Every beta is an estimate over a trailing window. It changes as the window rolls and changes *structurally* across regimes — a beta estimated in a calm bull market is not the beta that applies in a crash. Read “ $\beta = 0.60$ ” as noisy and conditional, not as a constant.
- **R^2 is frequently low, and we report it for that reason.** The demo book's R^2 of 43% means the market factor explains under half its variance; the rest is sector and stock-specific, and the beta says nothing about it. NVDA's two-factor fit is 54% and many names are worse. A beta shown without its R^2 invites more confidence than the estimate supports, which is why the two are always displayed together.
- **Correlation of daily returns understates tail dependence.** Pearson correlation describes *average* co-movement in normal conditions. It systematically understates how positions move together in a crisis, when correlations spike toward one and diversification evaporates exactly when it is needed. A comforting 0.05 average is not a promise about the next crash.
- **Language models are fluent, confident, and sometimes wrong.** Evidence-grounding and no-hallucination rules reduce this risk; they do not eliminate it. Confident tone is not evidence of correctness. Relatedly, **the debate is not a forecast and its confidence is not calibrated** — neither the vigour of an argument nor the Judge's verdict carries a calibrated probability.
- **The performance chart is a hypothetical backtest with composition bias.** It is a constant-current-weight, daily-rebalanced reconstruction using today's holdings projected backward, so it embeds look-ahead and composition bias: it assumes the investor held exactly today's book, rebalanced daily and frictionlessly, throughout the past. No transaction costs, spreads, market impact or taxes are modelled anywhere in the system.
- **This is not investment advice.** RoboSmart is an educational tool. It does not know the user's circumstances, constraints or objectives, and nothing it outputs is a recommendation to buy, sell or hold any security.

6. Future work

Realistic extensions within the same architecture, not aspirational rewrites:

- **Rolling betas with confidence intervals** — report betas as rolling series with standard errors so instability is visible rather than asserted in prose.
- **Programmatic grounding checks** — the prompts require every claim to cite a number, but nothing yet *verifies* that the cited figure appears in the context. A substring/numeric check over agent output would turn a prompt-level promise into an enforced guarantee, and would yield a measurable groundedness rate.
- **Cost and token accounting** — surface per-analysis spend in the UI and cap it per session; the machinery currently makes no measurement of what a debate costs.
- **A better-sourced news layer** — a timestamped, ticker-tagged provider would address the coverage limit that §5 can only narrow, not solve.
- **Risk-adjusted metrics** — a risk-free rate would unlock Sharpe and Sortino; historical VaR/CVaR would extend the tail-risk picture the correlation matrix currently only hints at.